Semantic and temporal review - consolidated V2

DeepSeek 4810 + CodeLlama 4309 | original figures retained; new discussion and 4309 analyses appended

Purpose

An internal evidence review for choosing what belongs in the paper and slides. This is not a finished manuscript narrative. Existing results, counterexamples and limitations are intentionally retained.

What is new

Comparison of all 27 changed CodeLlama target outputs with six shuffled-direction arms; structural hypotheses on natural base/merged texts; temporal localization across 50 regressions; return-newline preference; eight activation examples; all seven return-only cases, including unchanged and worsened implementations.

How to read older pages

The first 14 original pages are preserved without modification. Where later pages add nuance, use the newer analysis: return-newline preference is not error specificity, and continuous steering does not establish mediation by late natural activation.

Data scope

DeepSeek: 119 contamination improvements; CodeLlama: 50 logic/runtime regressions. Main report doses are 0 to 5. No new inference, generation, training or benchmark evaluation was performed for this compilation.

Reading guide and reproducibility

All page ranges refer to this consolidated PDF.

Section	Pages
Original visual review (preserved)	3-16
Updated findings and complete change tables	17-21
Exploratory behavior hypothesis figures	22-24
Uniform target / shuffle comparison	25-25
Temporal regions, return-newline statistics and eight examples	26-28
Code before / after and unchanged counterexamples	29-31

All 50 annotated programs

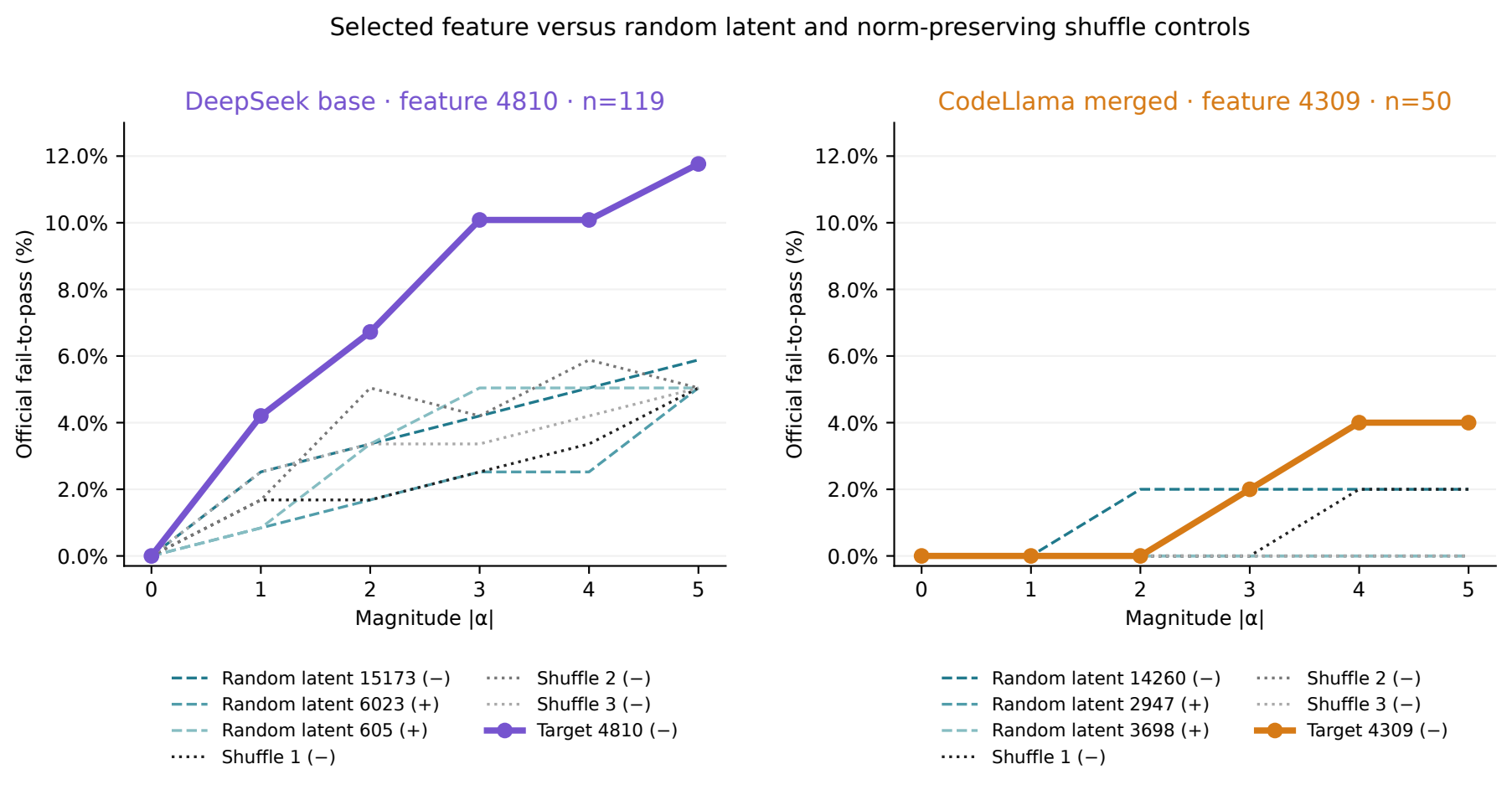
reports/CODELLAMA_4309_ALL_50_ANNOTATED_PTBR.md contains complete code with => / <= error regions, * first-activation marker, original line numbers, rationale and evaluator evidence.

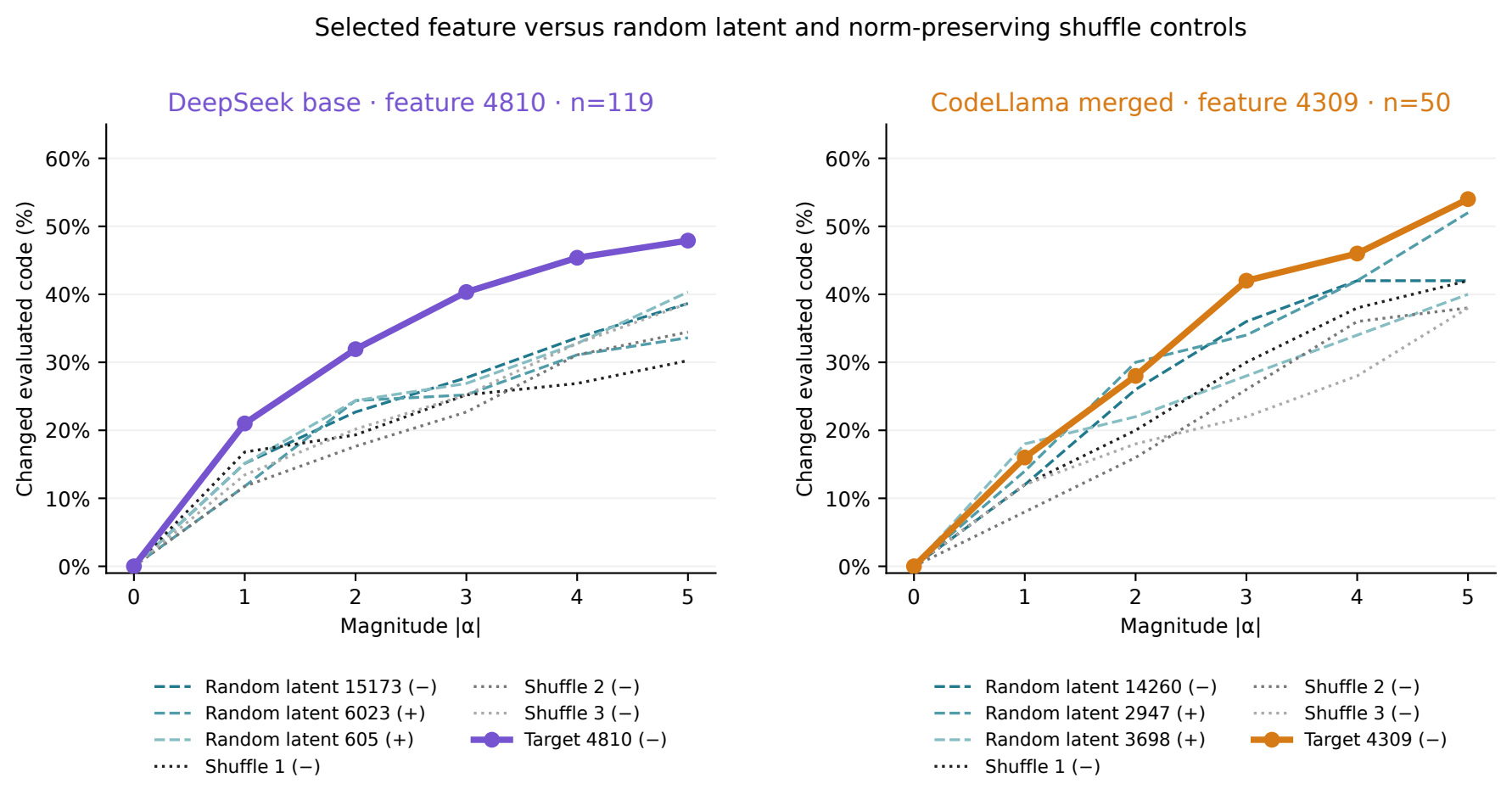
Natural capture versus intervention baseline

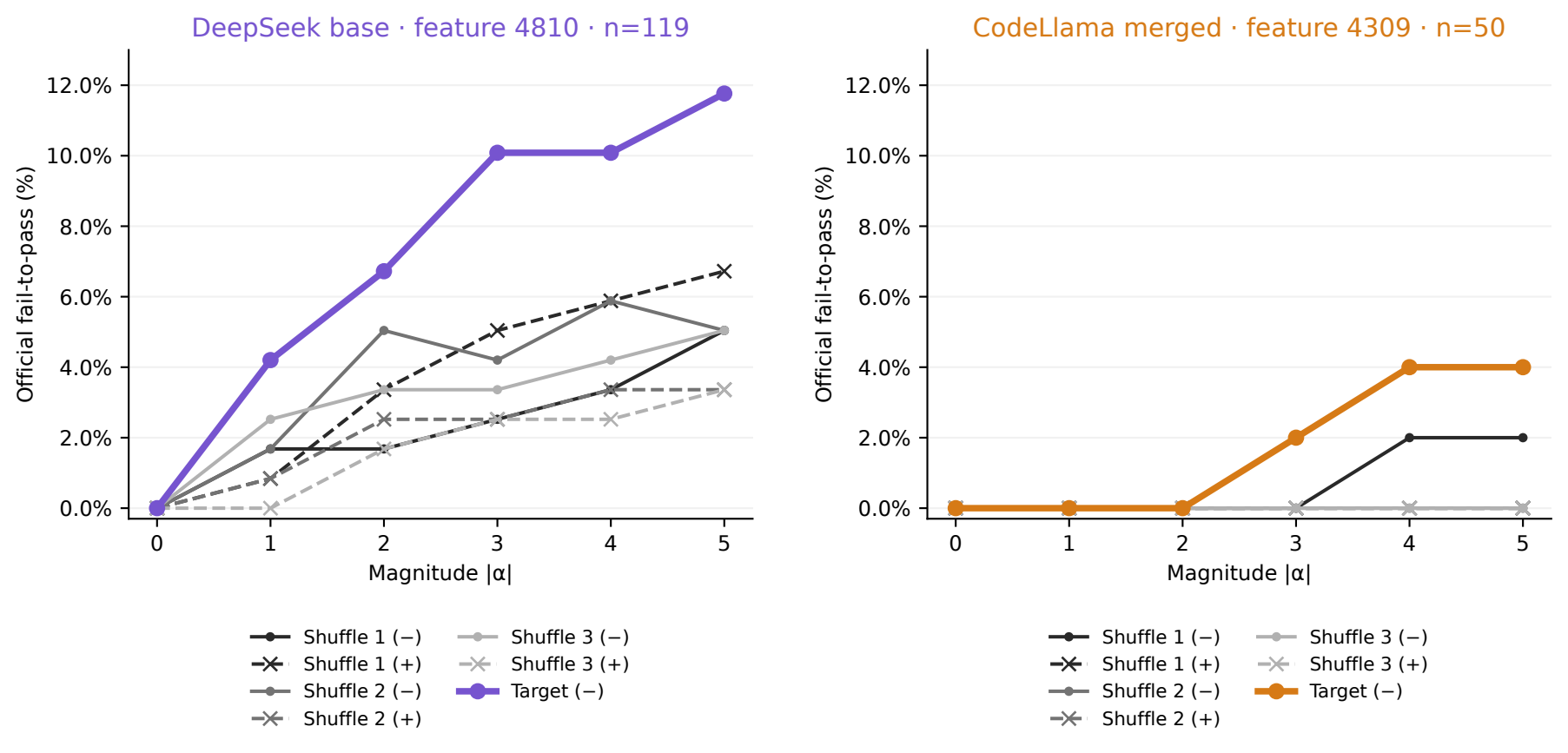
Frozen captured code differs from currently re-extracted alpha-zero code in some cases. Activation anchors are aligned by exact matching blocks. Offline shared-latent activation is not an online per-model measurement; prompt activations were excluded.

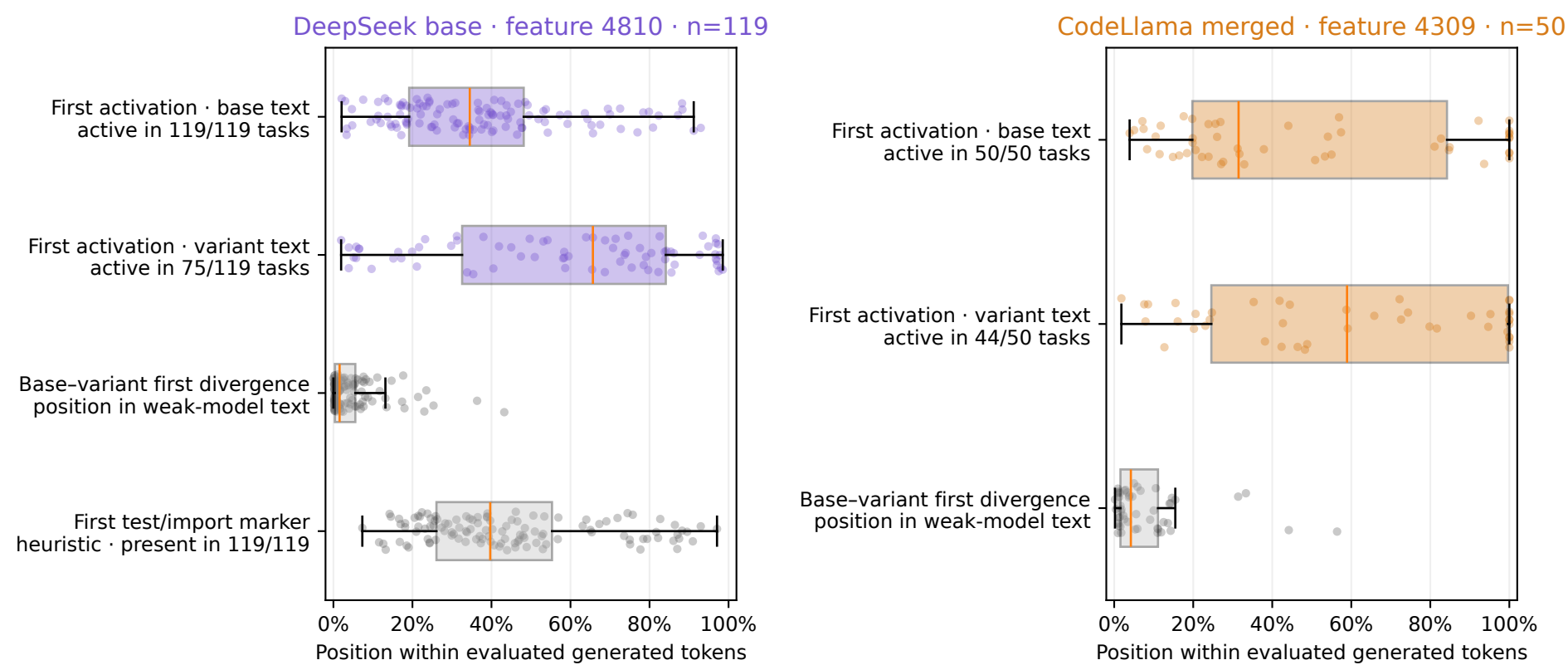
Controls and experimental status

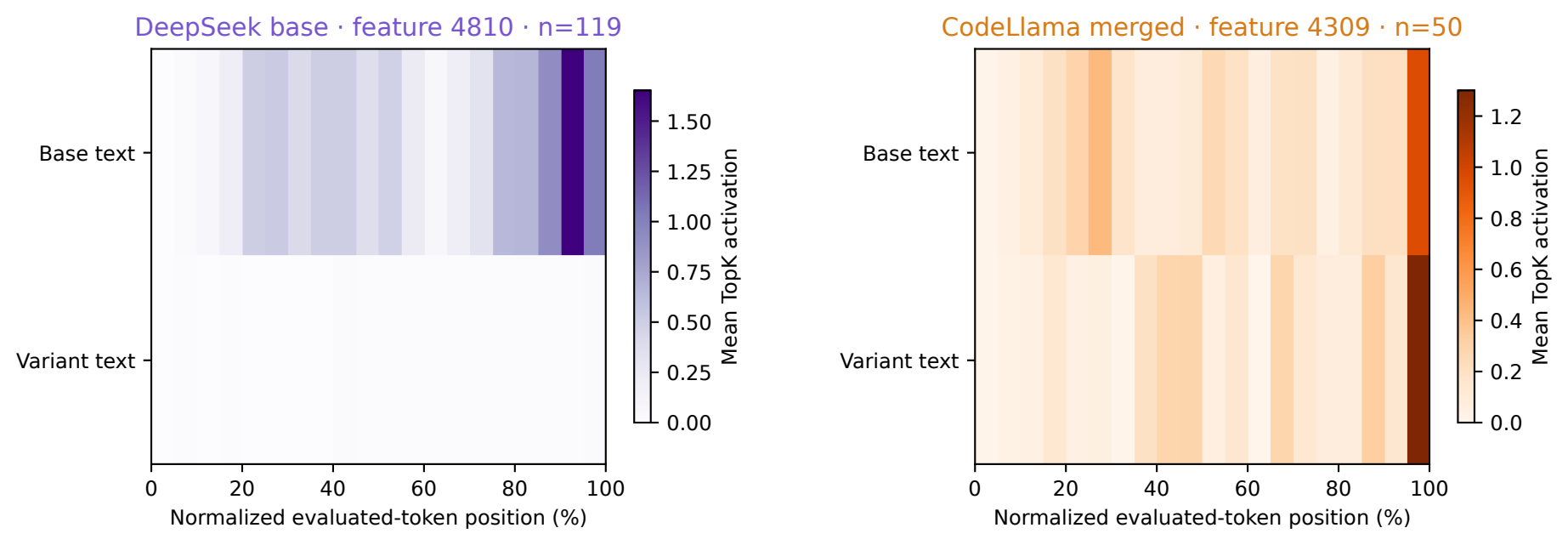
Three negative shuffled arms form the primary same-sign comparison; positive arms are sensitivity checks. Random CrossCoder latents are distinct from shuffled decoder coordinates. Current interpretations are exploratory on selected cohorts, without independent held-out validation.

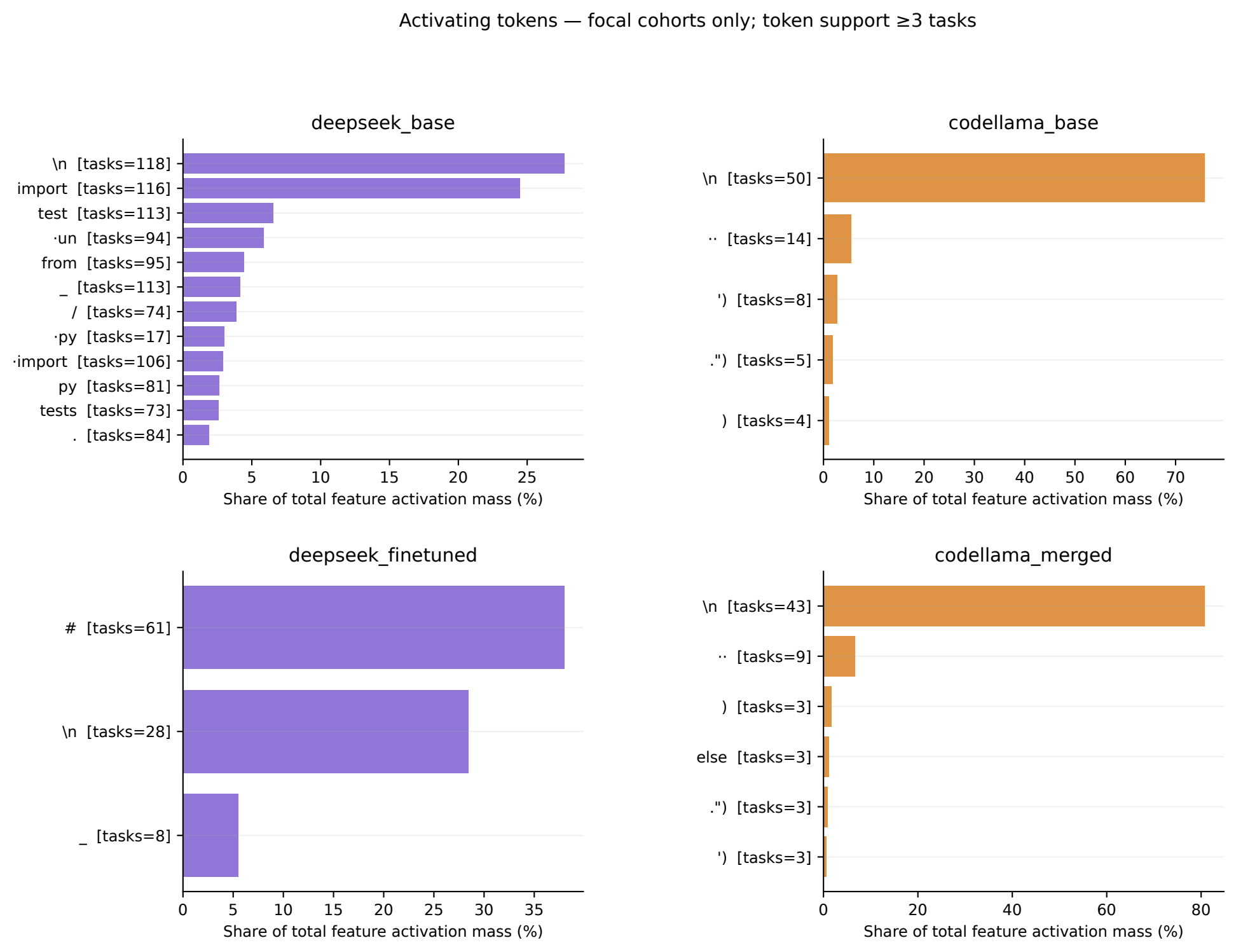

Selected feature versus random latent and norm-preserving shuffle controls
DeepSeek base · feature 4810 · n=119
CodeLlama merged · feature 4309 · n=50
Official fail-to-pass (%)
12.0%
10.0%
8.0%
6.0%
4.0%
2.0%
0.0%
0
1
2
3
4
5
Magnitude |α|
Random latent 15173 (−)
Random latent 6023 (+)
Random latent 605 (+)
Shuffle 1 (−)
Shuffle 2 (−)
Shuffle 3 (−)
Target 4810 (−)
Random latent 14260 (−)
Random latent 2947 (+)
Random latent 3698 (+)
Shuffle 1 (−)
Shuffle 2 (−)
Shuffle 3 (−)
Target 4309 (−)

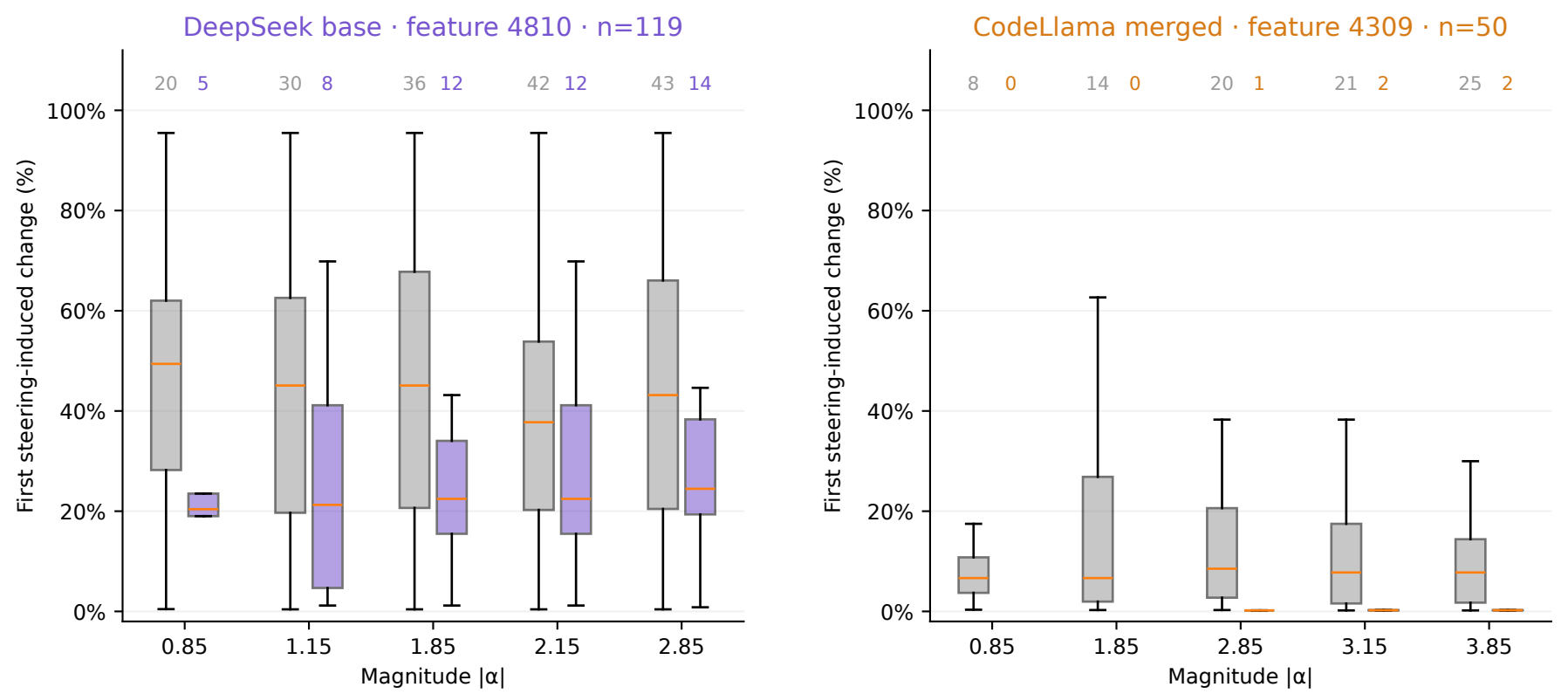

Selected feature versus random latent and norm-preserving shuffle controls
DeepSeek base · feature 4810 · n=119
CodeLlama merged · feature 4309 · n=50
Changed evaluated code (%)
Magnitude |α|
0%
10%
20%
30%
40%
50%
60%
0
1
2
3
4
5
Random latent 15173 (−)
Random latent 6023 (+)
Random latent 605 (+)
Shuffle 1 (−)
Shuffle 2 (−)
Shuffle 3 (−)
Target 4810 (−)
Random latent 14260 (−)
Random latent 2947 (+)
Random latent 3698 (+)
Shuffle 1 (−)
Shuffle 2 (−)
Shuffle 3 (−)
Target 4309 (−)


DeepSeek base · feature 4810 · n=119
CodeLlama merged · feature 4309 · n=50
Official fail-to-pass (%)
12.0%
10.0%
8.0%
6.0%
4.0%
2.0%
0.0%
0
1
2
3
4
5
Magnitude |α|
Shuffle 1 (−)
Shuffle 1 (+)
Shuffle 2 (−)
Shuffle 2 (+)
Shuffle 3 (−)
Shuffle 3 (+)
Target (−)


DeepSeek base · feature 4810 · n=119
First activation · base text active in 119/119 tasks
First activation · variant text active in 75/119 tasks
Base–variant first divergence position in weak-model text
First test/import marker heuristic · present in 119/119
0%
20%
40%
60%
80%
100%
Position within evaluated generated tokens
CodeLlama merged · feature 4309 · n=50
First activation · base text active in 50/50 tasks
First activation · variant text active in 44/50 tasks
Base–variant first divergence position in weak-model text
0%
20%
40%
60%
80%
100%
Position within evaluated generated tokens


DeepSeek base · feature 4810 · n=119
Base text
Variant text
0
20
40
60
80
100
Normalized evaluated-token position (%)
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Mean TopK activation
CodeLlama merged · feature 4309 · n=50
Base text
Variant text
0
20
40
60
80
100
Normalized evaluated-token position (%)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Mean TopK activation


Activating tokens — focal cohorts only; token support ≥3 tasks
deepseek_base
\n [tasks=118]
import [tasks=116]
test [tasks=113]
·un [tasks=94]
from [tasks=95]
_ [tasks=113]
/ [tasks=74]
·py [tasks=17]
·import [tasks=106]
py [tasks=81]
tests [tasks=73]
. [tasks=84]
0 5 10 15 20 25
Share of total feature activation mass (%)
codellama_base
\n [tasks=50]
·· [tasks=14]
') [tasks=8]
.") [tasks=5]
) [tasks=4]
0 10 20 30 40 50 60 70
Share of total feature activation mass (%)
deepseek_finetuned
[tasks=61]
\n [tasks=28]
_ [tasks=8]
0 5 10 15 20 25 30 35
Share of total feature activation mass (%)
codellama_merged
\n [tasks=43]
·· [tasks=9]
) [tasks=3]
else [tasks=3]
.") [tasks=3]
') [tasks=3]
0 20 40 60 80
Share of total feature activation mass (%)



Colored: corrected · gray: changed but still failing · numbers above: tasks · unchanged outputs omitted, not placed at 100%

Semantic case studies — corrected programs and changed-but-failing programs

DEEPSEEK · BigCodeBench/937 · α=−5

```
@@ -29,16 +29,5 @@
     return char_counts

-#test_task.py
-import unittest
-from task import task_func
-
-class TestTask(unittest.TestCase):
-    def test_task_func(self):
-        # Test case 1: Basic test
-        input_str = "Hello, World!"
-        expected_output = {'l': 3, 'o': 2, 'h': 1, 'e': 1, 'w': 1,
'r': 1, 'd': 1}
-        self.assertEqual(task_func(input_str), expected_output)
-
-        # Test case 2: Empty string
-        input_str = ""
-        expected_output = {}
+# Test the function
+input_str = "Hello, World!"
+print(task_func(input_str))
```

PASS: removed failing test-module import; solution body unchanged.
Baseline first activation: 40.5% · first code change: 40.5%

DEEPSEEK · BigCodeBench/475 · α=−5

```
@@ -67,5 +67,5 @@

     # Create a histogram of the data
-    ax =
data['dates'].dt.month.value_counts().sort_index().plot(kind='bar')
+    ax =
data['dates'].dt.month.value_counts().sort_index().plot(kind='bar',
color='blue', alpha=0.5)

     # Set the title and labels
@@ -77,7 +77,2 @@

     return ax
-
-#tests/test_task_func.py
-import unittest
-import pandas as pd
-from task_func import task_func
```

FAIL: test imports removed, but evaluator still reports missing locale.
Baseline first activation: 93.0% · first code change: 77.8%

CODELLAMA · BigCodeBench/490 · α=−5

```
@@ -26,7 +26,10 @@
     'Emma'
     """
-    return xmltodict.parse(s)
+    result = xmltodict.parse(s)
+    with open(file_path, 'w') as f:
+        json.dump(result, f)
+    return result

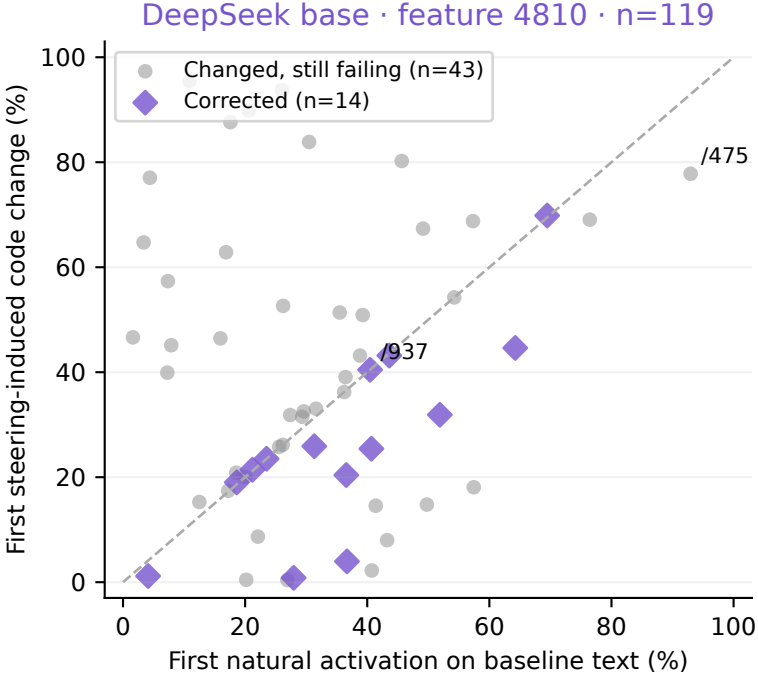
-def task_func2(s, file_path):
[Additional duplicate task_func_2 omitted from this excerpt.]
```

PASS: file-writing behavior restored; duplicate helper remains in output.
Baseline first activation: 2.0% · first code change: 0.2%

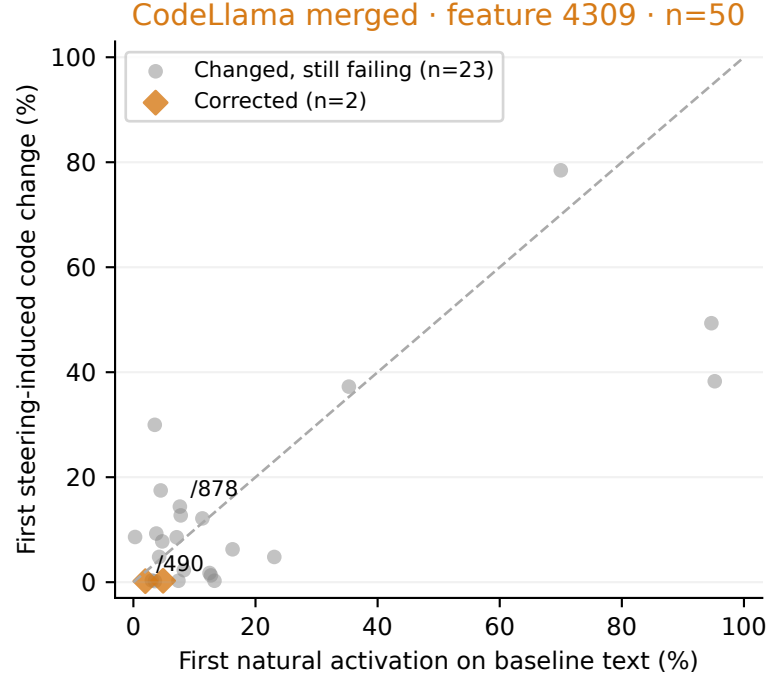
CODELLAMA · BigCodeBench/878 · α=−5

```
@@ -56,30 +56,22 @@
     if target is None or target not in data:
         raise ValueError("The target column name is not in the
DataFrame.")
+    if test_size is None or test_size < 0 or test_size > 1:
+        raise ValueError("The fractional size of the test set is
not between 0 and 1.")
     if random_state is None:
-        random_state = None
-    if test_size is None:
-        test_size = 0.2
-    if test_size > 1:
-        raise ValueError("The fractional size of the test set is
greater than 1.")
-    if test_size < 0:
-        raise ValueError("The fractional size of the test set is
less than 0.")
-    if test_size < 0.01:
-        raise ValueError("The fractional size of the test set is
less than 0.01.")
-    if test_size > 1:
-        raise ValueError("The fractional size of the test set is
greater than 1.")
-    if test_size < 0:
[Excerpt ends here; complete diff in CASE_INSPECTION.md]
```

FAIL: validation changes, but repeated checks remain and the task still fails.
Baseline first activation: 7.6% · first code change: 14.4%



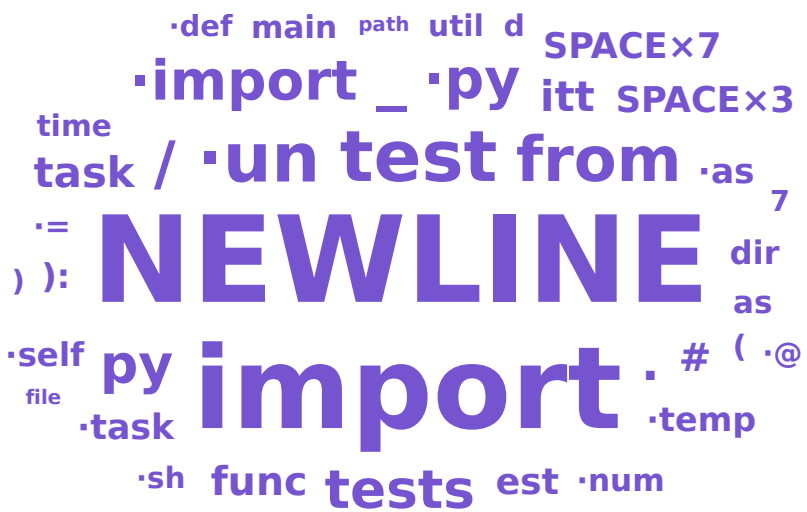
Changed outputs: naturally inactive=0; unaligned active anchor=0



Changed outputs: naturally inactive=1; unaligned active anchor=1

α=−5 · Below the diagonal: code changed before the feature first activated naturally. Steering here is unconditional.

Activating-token clouds — original focal-cohort texts

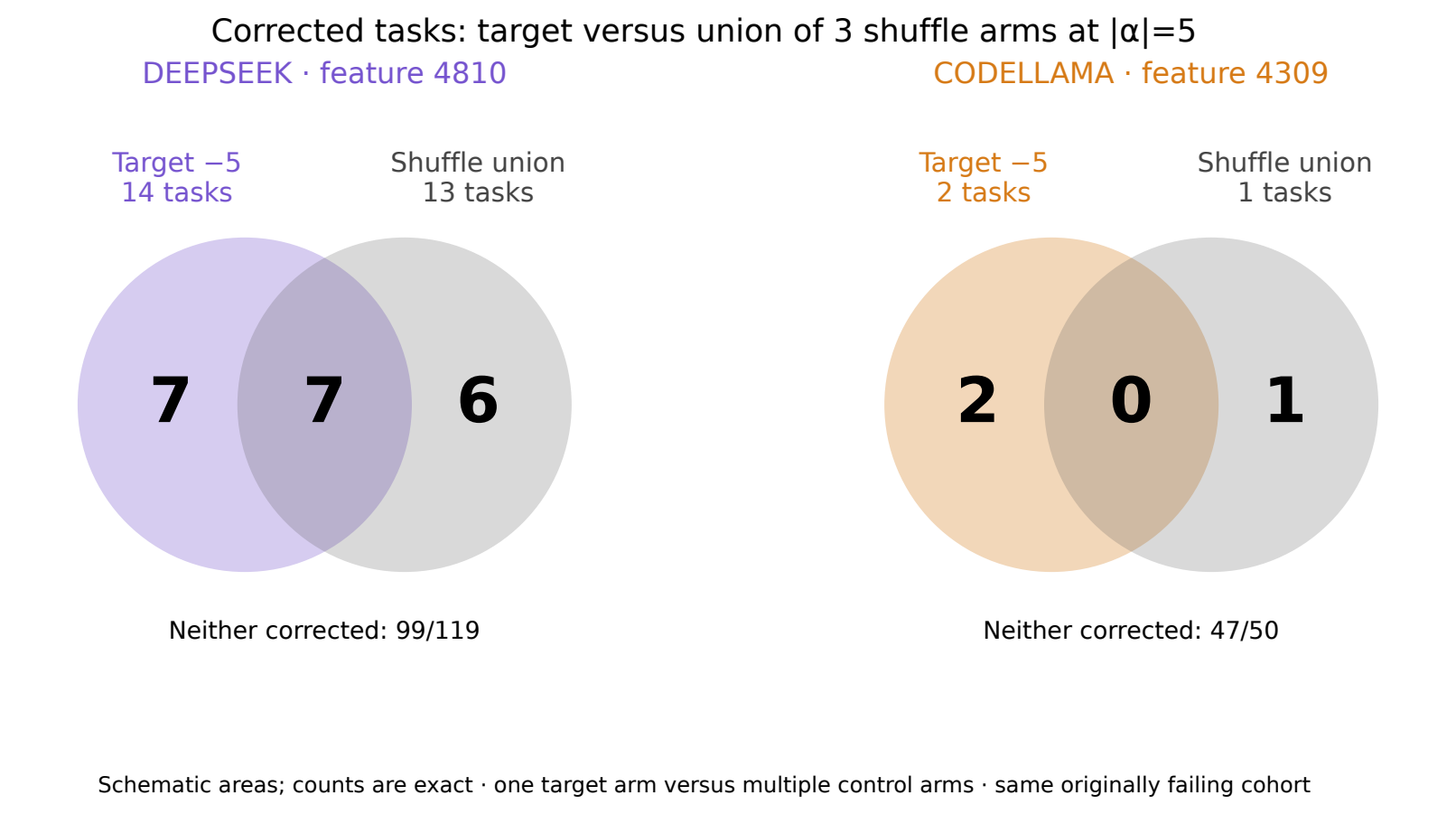






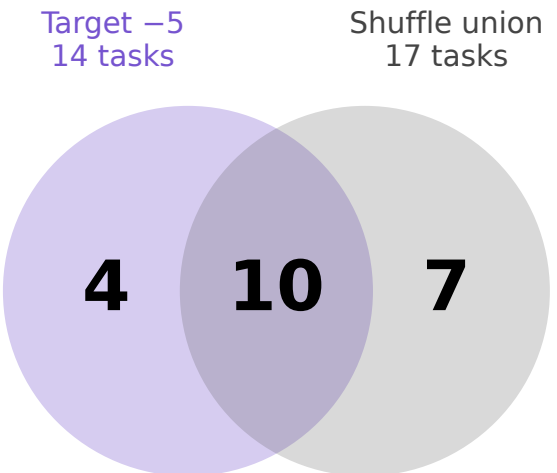


Font size increases with summed TopK activation · support ≥3 tasks · NEWLINE/SPACE are visible aliases, not words
119 DeepSeek tasks / 50 CodeLlama tasks · separate base/variant texts encoded in the shared same-text latent space


Corrected tasks: target versus union of 3 shuffle arms at |α|=5
DEEPSEEK · feature 4810
CODELLAMA · feature 4309
Target −5
14 tasks
Shuffle union
13 tasks
7
7
6
Neither corrected: 99/119
Target −5
2 tasks
Shuffle union
1 tasks
2
0
1
Neither corrected: 47/50
Schematic areas; counts are exact · one target arm versus multiple control arms · same originally failing cohort

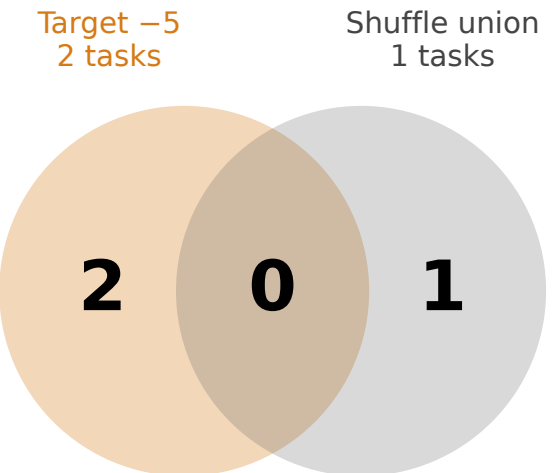
Corrected tasks: target versus union of 6 shuffle arms at |α|=5

DEEPSEEK · feature 4810



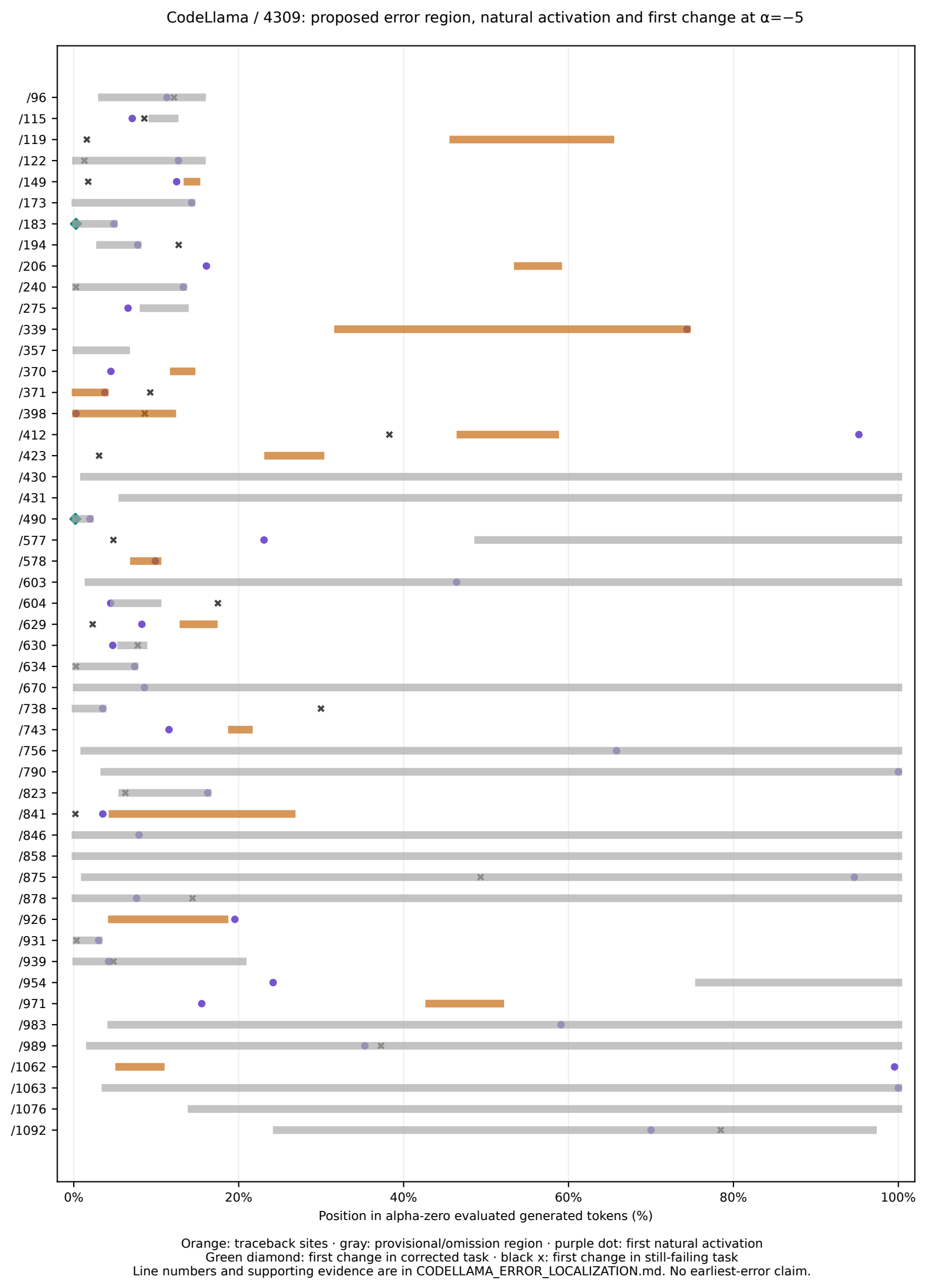
Neither corrected: 98/119

CODELLAMA · feature 4309



Neither corrected: 47/50

Schematic areas; counts are exact · one target arm versus multiple control arms · same originally failing cohort



Orange: traceback sites · gray: provisional/omission region · purple dot: first natural activation
Green diamond: first change in corrected task · black x: first change in still-failing task
Line numbers and supporting evidence are in CODELLAMA_ERROR_LOCALIZATION.md. No earliest-error claim.

Both CodeLlama successes: restoring required behavior in the main function

BigCodeBench/183 · merged baseline: FAIL

```
return HttpResponse(json.dumps(data), status=200)
```

BigCodeBench/183 · 4309 / α=−5: PASS

```
response = HttpResponse(data)
response['UUID'] = uuid.uuid4()
return response
```

BigCodeBench/490 · merged baseline: FAIL

```
return xmltodict.parse(s)
```

BigCodeBench/490 · 4309 / α=−5: PASS

```
result = xmltodict.parse(s)
with open(file_path, 'w') as f:
    json.dump(result, f)
return result
```

Main-function bodies only; full outputs still contain duplicate functions. Test assertions and natural activations are preserved in the accompanying report.

What differs from shuffled directions at magnitude five?

4309 alpha -5 versus three shuffled decoder directions at alpha -5 | same 50 failing merged tasks

Observation	4309	Shuffle 1	Shuffle 2	Shuffle 3
Changed programs	27	21	19	19
Changed main-body AST	24	17	14	17
Official fail-to-pass	2	1	0	0
Fewer failures, still fail	2	2	3	2
More failures	4	5	3	3
Return-only expansion	3	0	0	0
Explicit value-return lost	6	0	1	1

Interpretation

More edits or longer programs are not specific to the target. Its two official corrections (/183 and /490) are absent from all six shuffle arms; partial improvements also occur under controls. Counts are exploratory, not an independent confirmatory comparison.

All 27 changed target programs

Uniform saved-evaluation / AST comparison at alpha -5. Test-count direction is descriptive, not a semantic correctness score.

Task	Outcome	Main AST changed	Failing tests	First activation vs region
/96	FAIL	yes	4 -> 4	within
/115	FAIL	yes	3 -> 3	before
/119	FAIL	yes	5 -> 5	unavailable
/122	FAIL	yes	4 -> 4	within
/149	FAIL	yes	5 -> 5	before
/183	PASS	yes	5 -> 0	within
/194	FAIL	no	5 -> 5	within
/240	FAIL	yes	2 -> 5	within
/371	FAIL	no	5 -> 5	within
/398	FAIL	yes	3 -> 3	within
/412	FAIL	yes	3 -> 3	after
/423	FAIL	yes	5 -> 3	unavailable
/490	PASS	yes	1 -> 0	within
/577	FAIL	yes	5 -> 5	before
/604	FAIL	yes	2 -> 2	before
/629	FAIL	yes	5 -> 5	before
/630	FAIL	yes	5 -> 5	before
/634	FAIL	yes	5 -> 5	within
/738	FAIL	no	5 -> 5	within
/823	FAIL	yes	5 -> 5	within
/841	FAIL	yes	4 -> 5	before
/875	FAIL	yes	2 -> 7	within
/878	FAIL	yes	7 -> 5	within
/931	FAIL	yes	5 -> 7	within
/939	FAIL	yes	4 -> 4	within
/989	FAIL	yes	7 -> 7	within
/1092	FAIL	yes	3 -> 3	within

Of 25 changed but still-failing outputs: 2 have fewer recorded failures, 4 more, 19 the same count.

Does natural activation explain a general omission pattern?

Original base-generated versus merged-generated texts | same shared CrossCoder feature | 50 regression pairs

Structural indicator	Passing base /50	Failing merged /50
Return-only main function	1	7
No explicit value return	3	17
Potential fallthrough (proxy)	4	23
Mean activation, task average	0.1997	0.1979

Outside the two discovery examples

Among the other 48 tasks, only four newly become return-only. Their paired activation contrast versus the other 44 has E/V +0.015 (mean), -0.527 (max), +0.841 (active fraction). These results do not strongly support a general incomplete-implementation detector; the positive group is small.

Scope and confounding

The 48-task subset is not an independent held-out dataset. Mean activation correlates with newline density (Spearman about 0.35). Return-only and no-value-return are not automatic failure labels: legitimate passing implementations can have either structure.

All seven failing return-only implementations

Feature 4309, continuous steering alpha -5 | expansion does not imply correction

Task	AST statements	Observed change	Outcome	Natural first activation
/183	1 -> 3	Expanded	PASS	within return region
/357	1 -> 1	Main body unchanged	FAIL	inactive
/371	1 -> 1	Main body unchanged	FAIL	within return region
/490	1 -> 4	Expanded	PASS	within return region
/634	1 -> 35	Expanded	FAIL	within return region
/738	1 -> 1	Main body unchanged	FAIL	within return region
/931	1 -> 0	Return removed	FAIL	within return region

Three expansions, two repairs

/183 restores response handling and UUID header; /490 restores JSON persistence. /634 expands into repetitive chain assignments and remains wrong.

Three unchanged bodies and one deterioration

/357 is unchanged entirely; /371 and /738 change outside the main body only. /931 loses its return and retains no executable body beyond the docstring.

What the temporal evidence does - and does not - establish

Natural activation is not the same thing as a continuous additive intervention

Observed timing

First mapped activation: 13 before the localized region, 27 within it, 3 after, 7 unavailable (6 inactive in captured code, 1 unaligned). Regions mix runtime execution sites, provisional lines/intervals and omission regions; they are not validated first semantic errors.

The corrected tasks

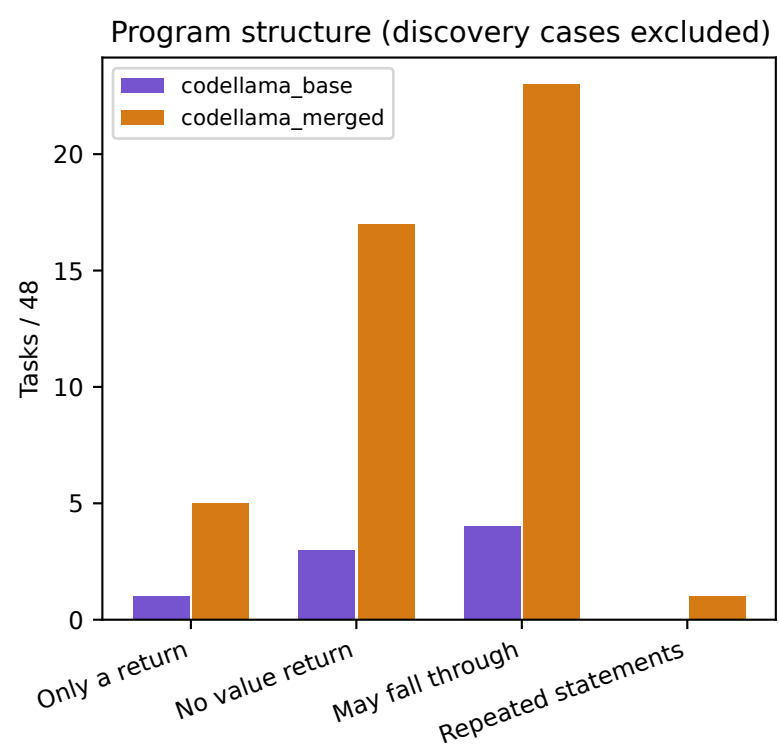
In /183 and /490 the first natural activation is the newline after the already-emitted return expression. Continuous steering changes the earlier return-versus-assignment decision. The direction has causal leverage, but these observations do not establish that natural activation at that newline caused the original omission.

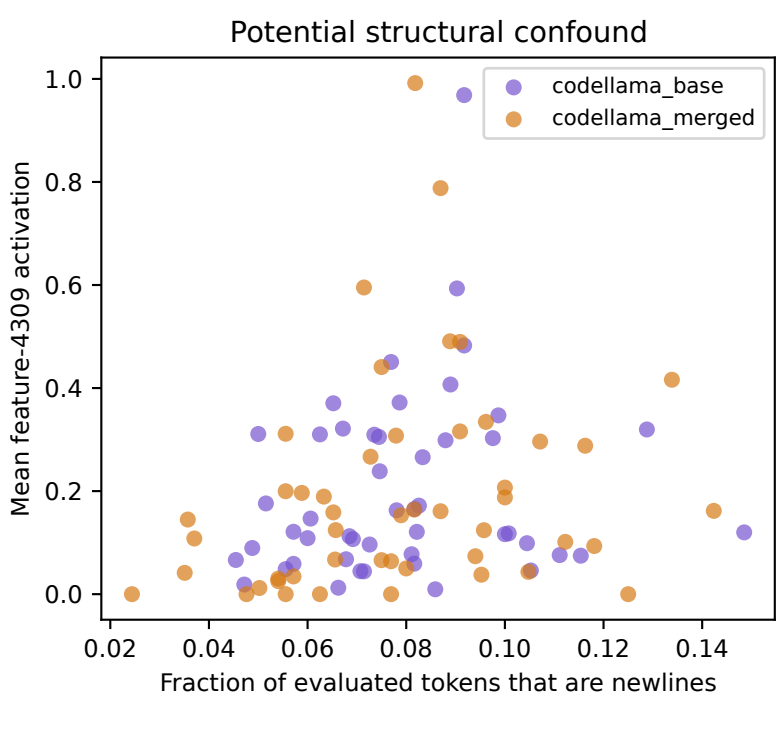
The return-newline preference

Activation occurs on 49/51 return-ending newline tokens in merged text (96.1%), versus 92/327 other newline tokens (28.1%). Passing base text also shows this preference: 48/53 (90.6%) versus 127/403 (31.5%). This is not an error-specific detector.

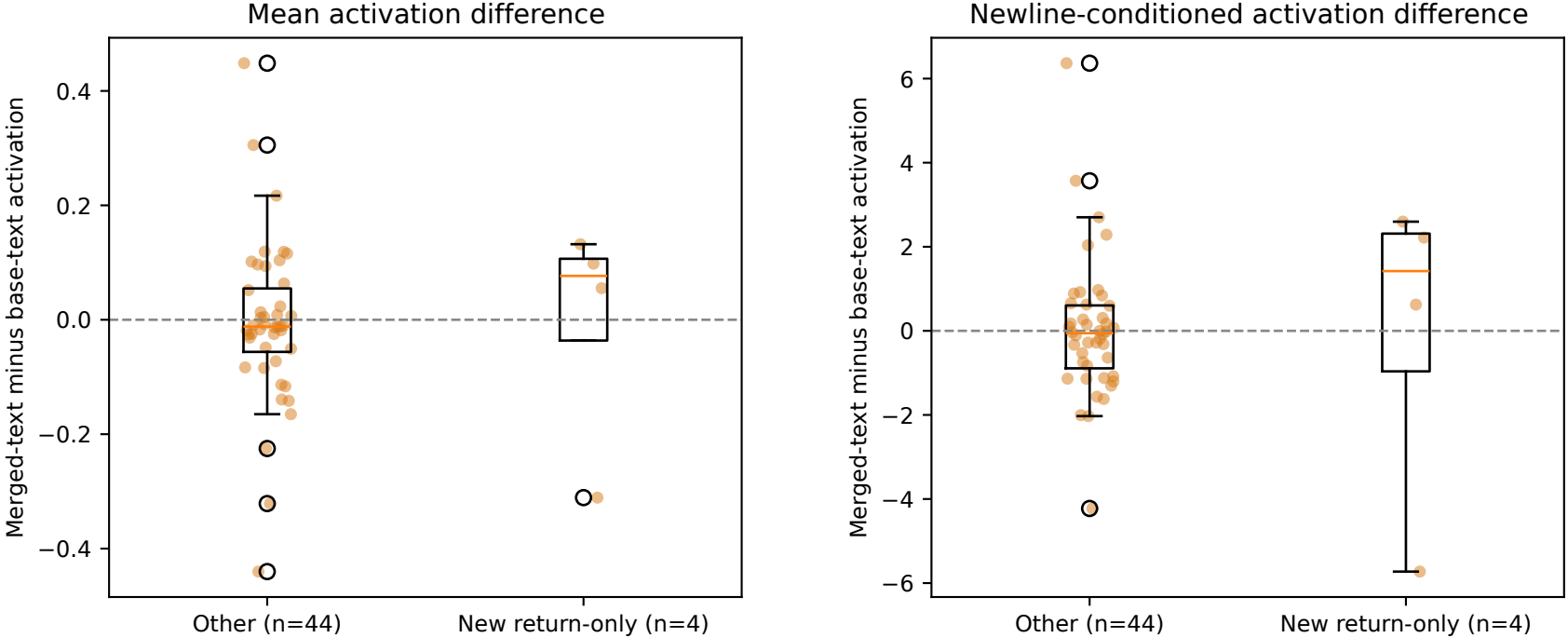
The most defensible reading

A shared sparse direction associated with line/return boundaries can cause specific implementation repairs, but also neutral and harmful rewrites. Neither a universal error detector nor a general natural mediator has been established.

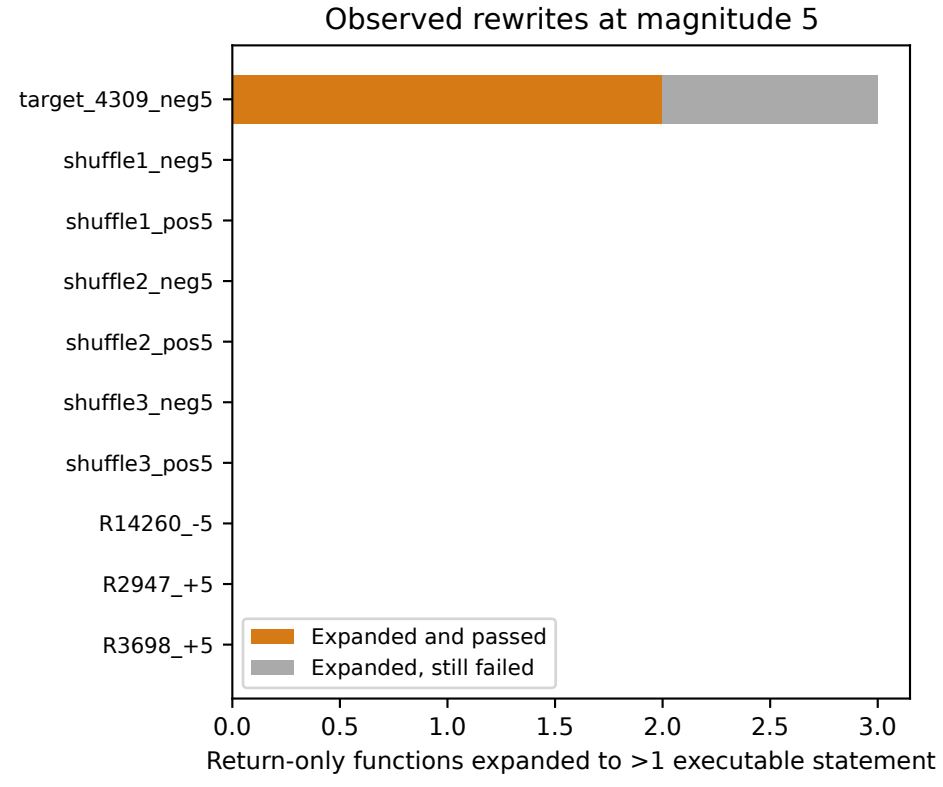


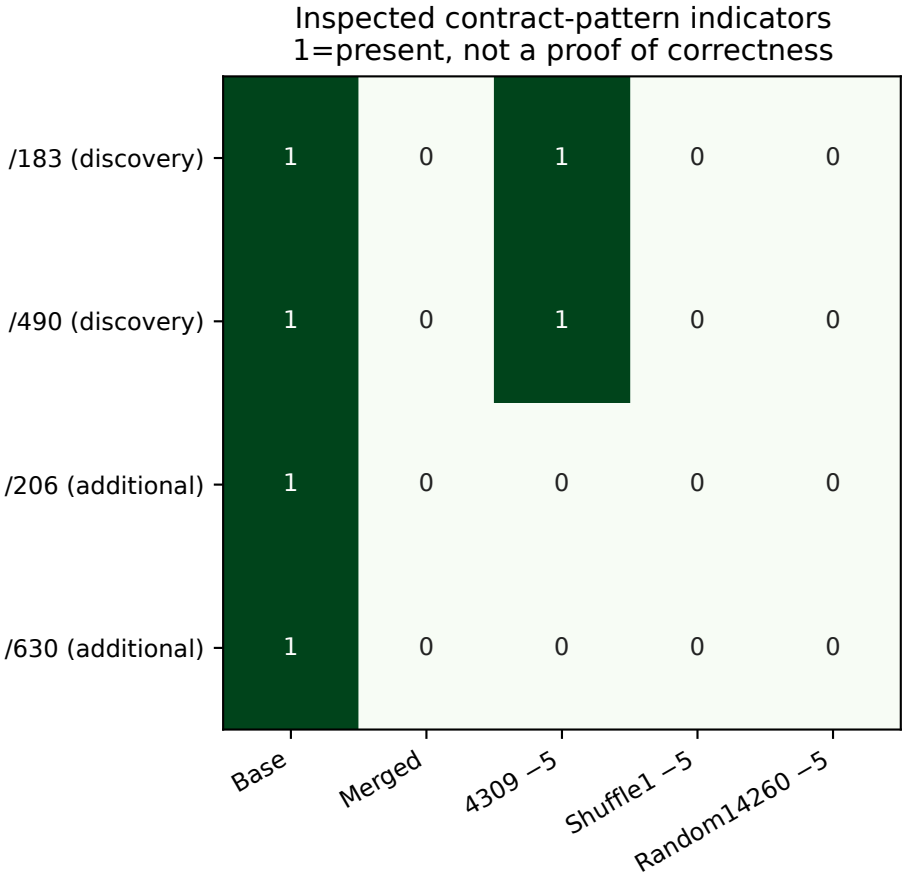


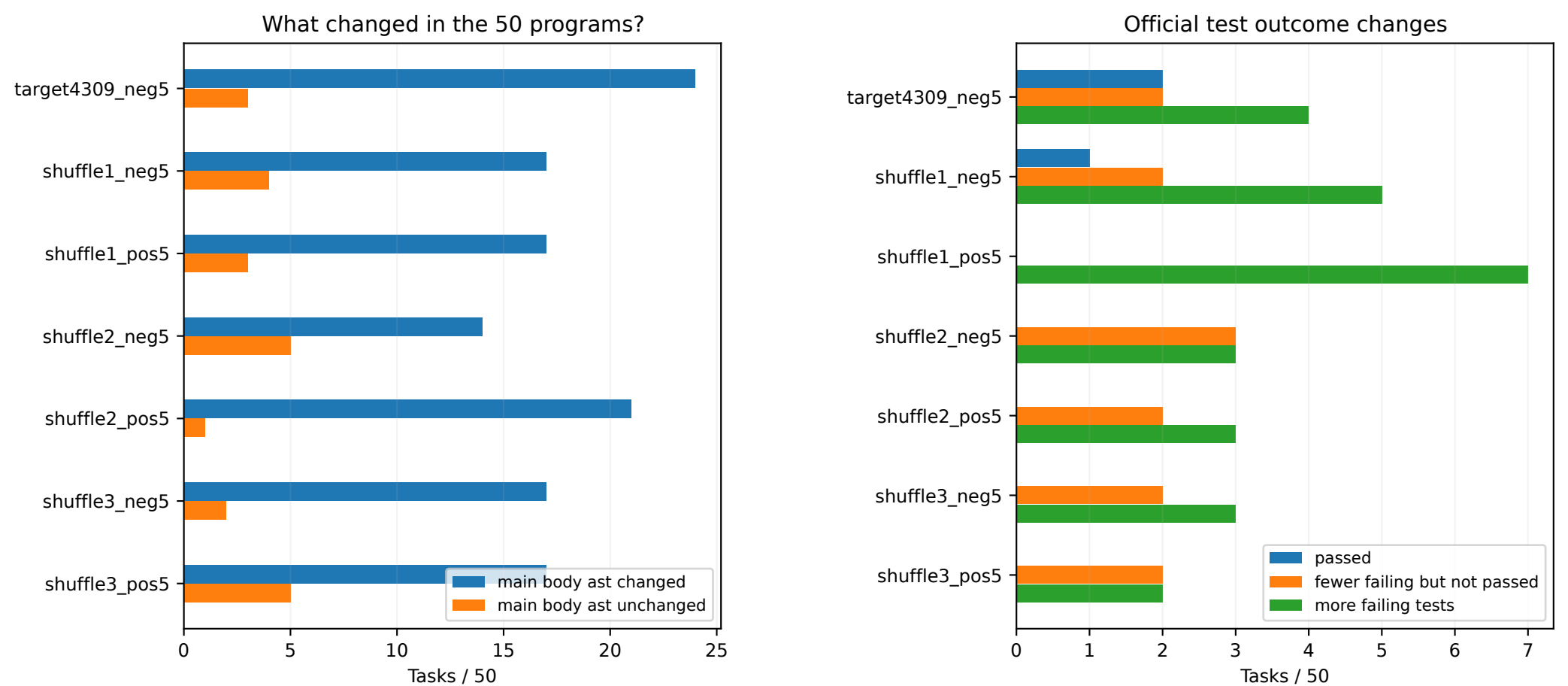
Exploratory within-cohort diagnostics · AST indicators are not a correctness oracle or complete control-flow analysis.



The 48 non-discovery tasks still belong to the original exploratory cohort; they are not a fresh validation set.

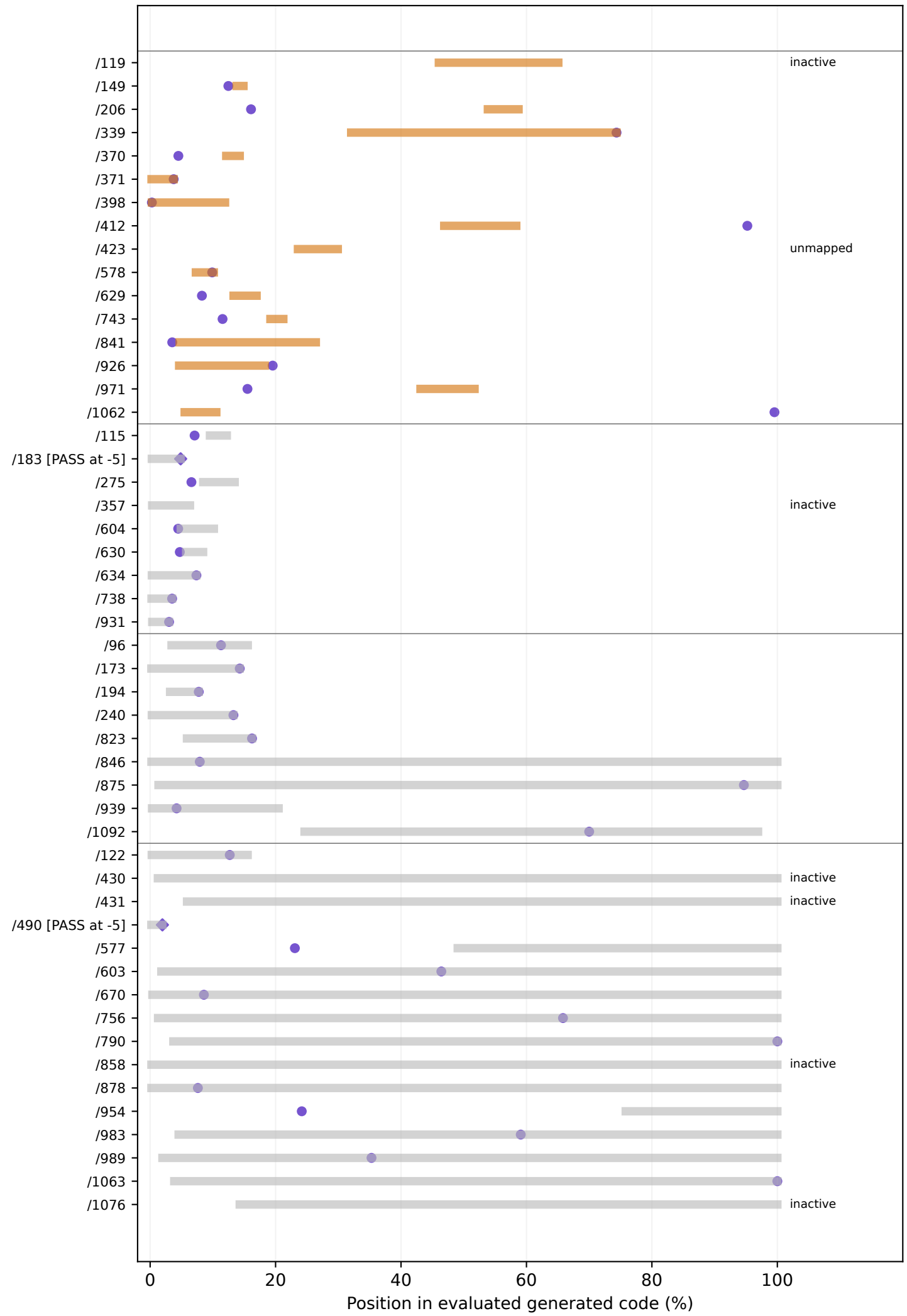

Observed rewrites at magnitude 5
target_4309_neg5
shuffle1_neg5
shuffle1_pos5
shuffle2_neg5
shuffle2_pos5
shuffle3_neg5
shuffle3_pos5
R14260_-5
R2947_+5
R3698_+5
Expanded and passed
Expanded, still failed
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Return-only functions expanded to >1 executable statement

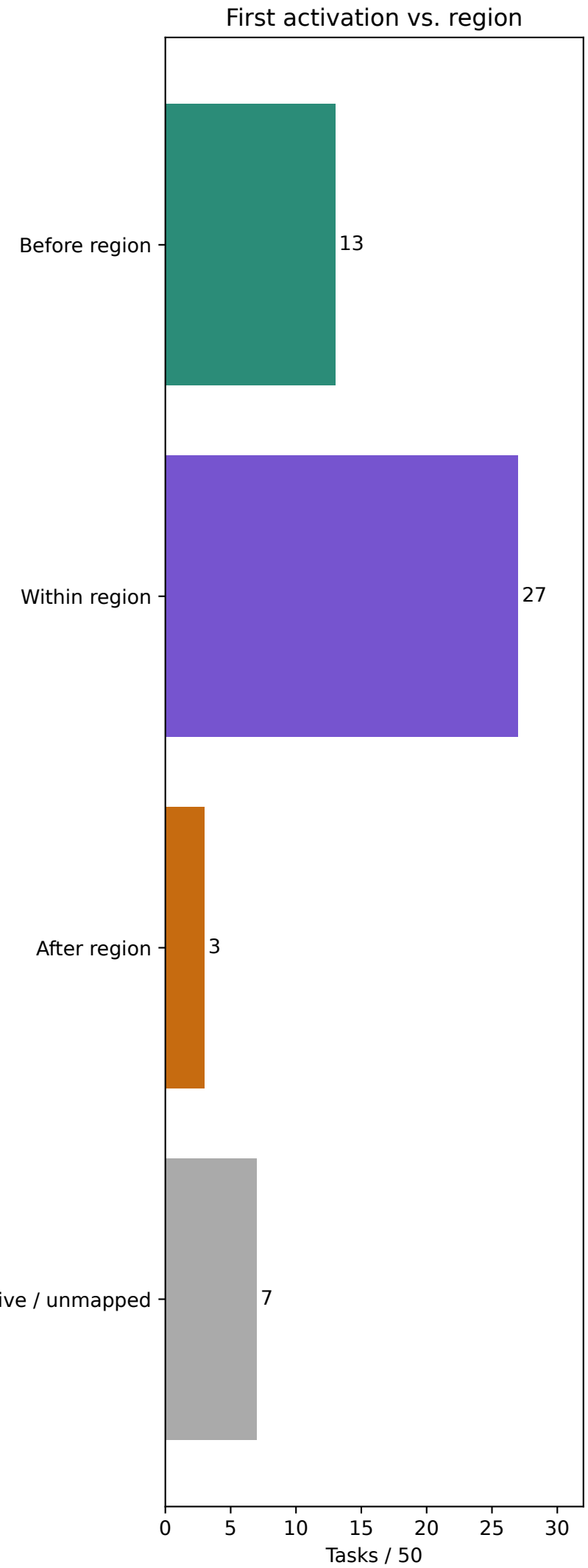

Inspected contract-pattern indicators
1=present, not a proof of correctness
/183 (discovery)
/490 (discovery)
/206 (additional)
/630 (additional)
1 0 1 0 0
1 0 1 0 0
1 0 0 0 0
1 0 0 0 0
Base
Merged
4309 −5
Shuffle1 −5
Random14260 −5



Same task cohort and magnitude · categories are not all mutually exclusive · failing/blocked-test counts are not semantic proof.

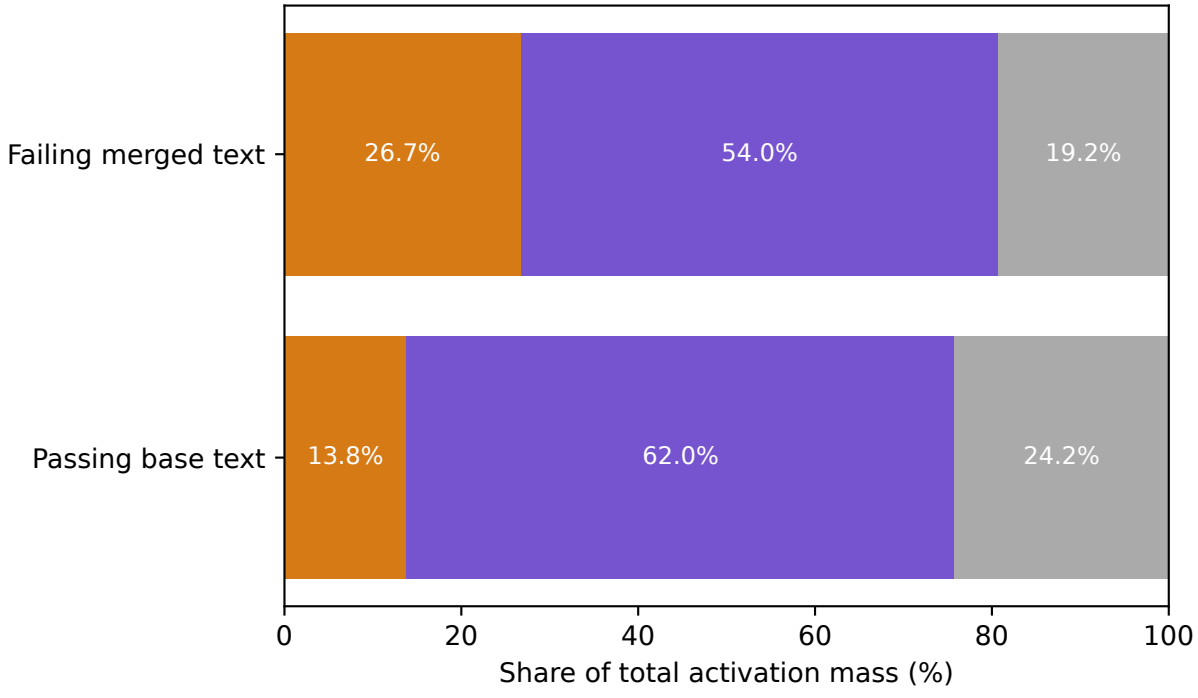
4309 timing in 50 failing merged outputs

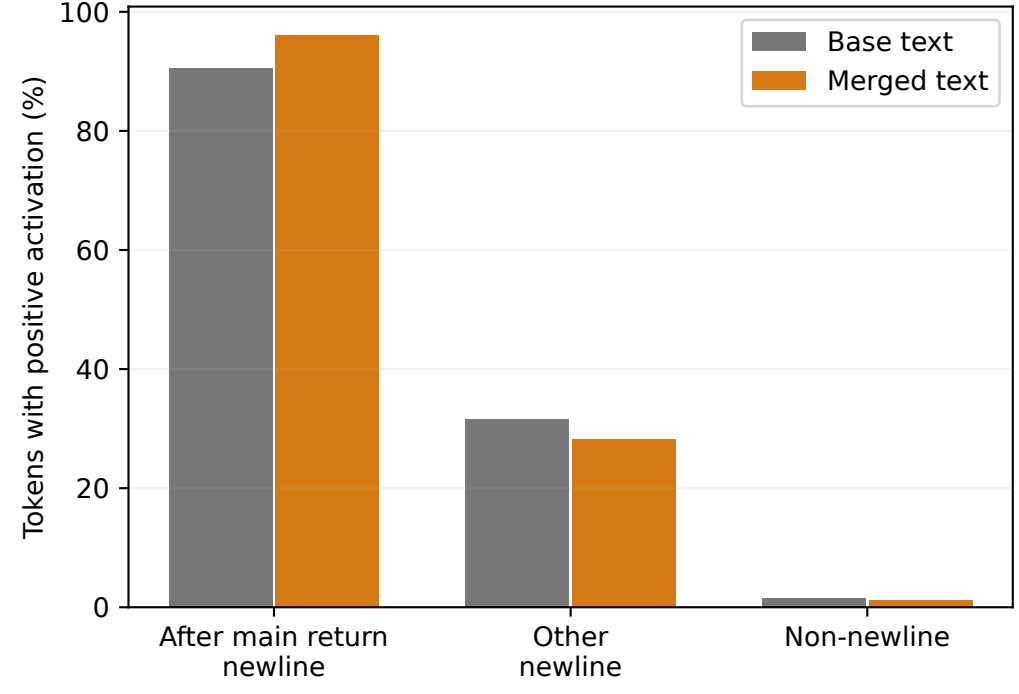




Purple: first natural activation · region: gray (proposed), orange (runtime site) · diamonds: two corrected tasks
Rows grouped: runtime sites / proposed lines / proposed intervals / omission regions. These are NOT validated error onsets.

Does 4309 prefer return-line boundaries?





50 task pairs · same shared feature on different generated texts · pooled descriptive statistics, not independent token-level tests.

Natural first-activation examples: return boundaries and counterexamples

BigCodeBench/183 | PASS at alpha -5

```
22      True
23      """
24      return HttpResponse(json.dumps(data), status=200)* [newline]
```

First activation: newline_after_main_return | z=4.552
Timing vs. region: same_token_or_within_region

BigCodeBench/490 | PASS at alpha -5

```
26      'Emma'
27      """
28      return xmltodict.parse(s)* [newline]
```

First activation: newline_after_main_return | z=3.357
Timing vs. region: same_token_or_within_region

BigCodeBench/634 | FAIL at alpha -5

```
32      ModeResult(mode=array([1]), count=array([5]))
33      """
34      return stats.mode(itertools.chain(input_list, input_list, input_list,
input_list))* [newline]
```

First activation: newline_after_main_return | z=4.055
Timing vs. region: same_token_or_within_region

BigCodeBench/115 | FAIL at alpha -5

```
30      """
31      if len(numbers) == 0:
32          raise ValueError('task_func() requires a non-empty list of
numbers')* [newline]
33      else:
```

First activation: other_newline | z=13.332
Timing vs. region: before

BigCodeBench/629 | FAIL at alpha -5

```
28    # Create output directory if it does not exist.
29     if not os.path.exists(output_dir):
30         os.mkdir(output_dir)* [newline]
31
```

First activation: other_newline | z=7.036
Timing vs. region: before

BigCodeBench/122 | FAIL at alpha -5

```
29      my_list = np.array(my_list)
30      random_number = random.randint(0, 100)
31      my_list = np.append(my_list, random_number)* [newline]
32      return np.array(my_list)
```

First activation: other_newline | z=1.814
Timing vs. region: same_token_or_within_region

BigCodeBench/206 | FAIL at alpha -5

```
36      """
37      if not os.path.exists(file_name):
38          raise FileNotFoundError(file_name*)
39      with open(file_name, 'r', newline='') as csvfile:
```

First activation: non_newline | z=9.161
Timing vs. region: before

BigCodeBench/96 | FAIL at alpha -5

```
36      with open(csv_file, "r") as f:
37          data = f.read().split(csv_delimiter)
38          return [ (word, wordcount) for wordcount, word in
Counter(data).items() ]* [newline]
39          sorted(operator.itemgetter(1), operator.itemgetter(0))
```

First activation: newline_after_main_return | z=2.235
Timing vs. region: same_token_or_within_region

Exact frozen merged-text snippets; * marks the token start (newline shown explicitly).
Includes both corrected cases, failed expansion, before-region cases, other-newline and non-newline counterexamples.
Additional examples selected by ascending task ID within the stated category; not a representative random sample.

Both official corrections

Exact evaluated main-function bodies before and after continuous 4309 steering at alpha -5; snippets are not full programs.

BigCodeBench/183 | merged baseline FAIL

```
    return HttpResponse(json.dumps(data), status=200)
```

BigCodeBench/183 | steering PASS

```
    response = HttpResponse(data)
    response['UUID'] = uuid.uuid4()
    return response
```

BigCodeBench/490 | merged baseline FAIL

```
    return xmltodict.parse(s)
```

BigCodeBench/490 | steering PASS

```
    result = xmltodict.parse(s)
    with open(file_path, 'w') as f:
        json.dump(result, f)
    return result
```

Failed expansion and removal

Exact evaluated main-function bodies before and after continuous 4309 steering at alpha -5; snippets are not full programs.

BigCodeBench/634 | merged baseline FAIL

```
    return stats.mode(itertools.chain(input_list, input_list,
input_list, input_list))
```

BigCodeBench/634 | steering FAIL

```
    input_list = itertools.chain(input_list)
    input_list = itertools.chain(input_list)
    input_list = itertools.chain(input_list)
    input_list = itertools.chain(input_list)
    input_list = itertools.chain(input_list)
    input_list = itertools.chain(input_list)
    input_list = itertools.chain(input_list)
    input_list = itertools.chain(input_list)
    input_list = itertools.chain(input_list)
... [display excerpt; full source in companion report]
    input
```

BigCodeBench/931 | merged baseline FAIL

```
    return defaultdict(int, {})
```

BigCodeBench/931 | steering FAIL

```
[No executable statement after the docstring]
```

Return-only failures that did not expand

All three main-function bodies below remain exactly unchanged at alpha -5.

BigCodeBench/357 | entire evaluated code unchanged

```
        return np.linspace(-10, 10, 1000)
```

BigCodeBench/371 | changes outside main function only

```
        return pd.DataFrame(MinMaxScaler(l).transform(l))
```

BigCodeBench/738 | changes outside main function only

```
        return np.sqrt(np.var(np.array(L)))
```